

Randomized controlled trial of 2.5-cm versus 3.5-cm mesial temporal resection in temporal lobe epilepsy—part 1: intent-to-treat analysis

Johannes Schramm · T. N. Lehmann · J. Zentner ·
C. A. Mueller · J. Scorzin · R. Fimmers ·
H. J. Meencke · A. Schulze-Bonhage · C. E. Elger

Received: 8 September 2010 / Accepted: 25 November 2010 / Published online: 18 December 2010
© Springer-Verlag 2010

Abstract

Background Only one prospective randomized study on the extent of mesial resection in surgery for temporal lobe epilepsy (TLE) exists. This randomized controlled trial (RCT) examines whether 3.5-cm mesial resection is leading to a better seizure outcome than a 2.5-cm resection.

J. Schramm (✉) · C. A. Mueller · J. Scorzin
Department of Neurosurgery, University of Bonn,
Sigmund-Freud-Str. 25,
53105 Bonn, Germany
e-mail: Johannes.Schramm@ukb.uni-bonn.de

T. N. Lehmann
Department of Neurosurgery, Campus Virchow Klinikum,
Charité Berlin,
Berlin, Germany

J. Zentner
Department of Neurosurgery, University of Freiburg,
Freiburg, Germany

R. Fimmers
Institute of Medical Biometrics, Informatics and Epidemiology,
University of Bonn,
Bonn, Germany

H. J. Meencke
Department of Epileptology, Epilepsiezentrum Berlin
Brandenburg and von Bodelschwinghsche Anstalten,
Königin Elisabeth Hospital Herzberge,
Berlin, Germany

A. Schulze-Bonhage
Epilepsy Center, University Hospital of Freiburg,
Freiburg, Germany

C. E. Elger
Department of Epileptology, University of Bonn,
Bonn, Germany

Methods Three epilepsy surgery centers using similar MRI protocols, neuropsychological tests, and resection types for TLE surgery included 207 patients in a RCT with pre- and postoperative volumetrics. One hundred and four patients were randomized into a 2.5-cm resection group and 103 patients into a 3.5-cm resection group, i.e., an intended minimum resection length of 25 versus 35 mm for the hippocampus and parahippocampus. Primary outcome measure was seizure freedom Engel class I throughout the first year. The study was powered to detect a 20% difference in class I outcome. Seizure outcome was available for 207 patients, complete volumetric results for 179 patients. Outcome analysis was restricted to control of successful randomization and an intent-to-treat analysis of seizure outcome.

Results The mean true resection volumes were significantly different for the 2.5-cm and 3.5-cm resection groups; thus, the randomization was successful. Median resection volume in the 2.5-cm group was 72.86% of initial volume and 83.44% in the 3.5-cm group. At 1 year, seizure outcome Engel class I was 74% in the 2.5-cm and 72.8% in the 3.5-cm resection group.

Conclusions The primary intent-to-treat analysis did not show a different seizure freedom rate for the more posteriorly reaching 3.5-cm resection group. It appears possible that not maximal volume resection but adequate volume resection leads to good seizure freedom.

Keywords Temporal lobe epilepsy · Epilepsy surgery · Randomized controlled trial · Amygdalohippocampectomy

Introduction

Among the epilepsy syndromes amenable to surgical treatment, mesial temporal lobe epilepsy (MTLE) is the

one most frequently operated on. Seizure freedom rates between 60% and 70% have been reported in recent years. Recently, it could be demonstrated [40] in a prospective randomized trial that temporal lobe epilepsy surgery is superior to prolonged medical therapy.

The surgery for MTLE has for a long time been dominated by the question which structures have to be removed to achieve the optimal rate of seizure freedom. A number of studies have been devoted not only to the question which structures [10, 12, 19–21, 31] but also to what extent the mesial temporal lobe structures have to be resected to achieve the highest rates of seizure freedom. Since Niemeyer [29] proved that the isolated removal of mesial structures, i.e., the combined hippocampus–parahippocampus (selective amygdalohippocampectomy, SAH) is also able to achieve seizure freedom, attention has focused on these mesial structures. One focus of discussion is how far backward from the tip should these mesial structures be resected.

A number of retrospective studies trying to define positive predictors of good seizure outcome after temporal lobe resection (TLR) and SAH pointed out that the extent of mesial resection didn't seem to play a role [16, 18, 23, 32, 36, 38]; fewer studies came to the conclusion that a larger extent is correlated with a better outcome [6, 7, 28, 35, 42]. There are only four randomized studies in the literature which look at extent of resection, one comparing SAH with TLR [13], one comparing two approaches for SAH [22], one comparing different extents of neocortical removal in TLR [15], and only Wyler's study [42] randomizing for mesial resection extent. The latter demonstrated that in temporal lobe resection, the rate of seizure freedom is considerably better if a hippocampal resection reaching far backward is performed [42]. There was one major problem with this study, namely the lack of postoperative magnetic resonance imaging (MRI) control of extent of resection. This is important since several authors have pointed out [3, 17, 28, 30] that it is an absolute requirement to precisely assess the extent of resection by volumetric MRI. Thus, the results of purely retrospective studies or prospective studies without postoperative MRI verification of true extent of resection cannot be considered as first class evidence, as pointed out in a recent review [33].

The disadvantages of non-randomized studies as summarized by Wiebe et al. [39–41] are well-known, and therefore, the conclusion is justified that there is no reliable study available (i.e., randomized design, prospectively collected data, MRI volumetrically controlled resection extent) comparing different extents of mesial resection to support the claim that a larger mesial extent of resection, i.e., a more posteriorly reaching resection of the hippocampus–parahippocampal formation is leading to a better seizure outcome. It would be of great interest if this could be proven,

to systematically perform far posterior reaching resections, as the proven benefit of surgery [40] could be enhanced.

In summary, the following reasons seem to justify this prospective randomized study: only one randomized study tested this relationship and only a minority of studies was done with postop MRI volumetric control of true resection extent.

Vice versa, one would like to know whether a lesser extent of resection is associated with a lower rate of seizure freedom. To elucidate this question, we organized a prospective randomized controlled trial with volumetric assessment of resection extent, comparing two groups with planned different extents of mesial resection for mesial temporal lobe epilepsy: one with an intended minimum mesial resection of 25 mm from the tip of the temporal horn backwards and the other one with a mesial minimum resection of 35 mm from the tip of the temporal horn backwards.

Methods

Participants Patients in the three epilepsy surgery centers at the Universities of Bonn, Freiburg, and Berlin were recruited for this study between October 2002 and December 2005. The central data base was kept in one center, the other two centers had a local identical data base collecting all relevant patient data in a prospective fashion and regularly sending them to the central data base. In each center, patients with temporal lobe epilepsy were screened and evaluated, and if recruited for the study, were followed up by that center. The MRI studies were organized in each center following a common MRI protocol. MRI volumetry was done only centrally using an epilepsy-dedicated MRI protocol (for details, see Urbach et al.) [37], the key elements being 1-mm slice thickness and isotropic 1-mm³ voxels using manual tracings of six subregions and the ANALYZE software package (Biomedical imaging resource, Worcester, Minn.). A detailed description is given in Müller et al. 2007. Intrarater reliability is described in that study which was calculated with the intraclass correlation coefficient between 0.93 and 0.98 for the various subcompartments [27]. Patients were regularly seen in outpatient clinics, and the required data were collected and fed in the local data base regularly to be sent to the central data base.

Patients were eligible for this study if they suffered from drug-resistant temporal lobe epilepsy and presurgical evaluation had led to the recommendation of either a partial temporal lobe resection combined with amygdalohippocampectomy or a more restricted selected SAH. Only cases with mesial involvement were included, based on the presurgical evaluation. This included patients with mesial

temporal lobe epilepsy (TLE) from hippocampal sclerosis or without sclerosis, mesial TLE cases due to other lesions in the hippocampus and parahippocampal gyrus small enough to be randomized for a smaller and longer resection, and lesional or non-lesional TLE cases with a recommendation for partial lobe resection including amygdalo–hippocampo–parahippocampectomy. Aspects of patient selection, definition of TLE, and criteria during presurgical evaluation in the three centers can be found in several publications [4, 5, 8, 9, 14, 25, 26]. Patients needed to be at least 18 years old and able to understand the study plan and the procedure of informed consent. Drug resistance was defined as seizure duration of more than 2 years despite continuous treatment with at least two antiepileptic drugs at maximal tolerated doses. The further requirement was that the patients had to be available for a minimum follow-up in the city of surgery after 1 year and had to be willing to participate in the neuropsychological test after 1 year.

Patients were recruited between mid-October 2002 and end of December 2005. Preoperative volumetric MRIs were usually performed a few days before the surgery, postoperative volumetric MRIs were performed for a minimum of 3 months after the surgery, usually in the fourth month and in a few cases, in the fifth month. All patients were regularly followed up, the first time at 3 months after surgery, then at 6 months after surgery, and at 1 year after surgery. All patients, regardless of intervention group, were kept on the same postoperative antiepileptic drug regime as preoperatively. Neuropsychological testing was done during presurgical work-up and postoperative late neuropsychological testing was done 1 year after surgery, usually not later than 13 months. The neuropsychological analysis will be dealt with separately. Follow-up ended in December 2006. Volumetrics were finalized in 2007.

Presurgical evaluation was harmonized between the centers with all patients having video-EEG recordings, MRI evaluation using the same protocol in all three centers, and neuropsychological testing using the same test batteries in all three centers. Full details of neuropsychological testing have been published recently [14]. Within the study, two different resection types could be used (SAH and partial lobectomy). Partial lobectomies were performed in two variants either as two-third anterior lobectomy (TLR) or as one-third anterior lobectomy (i.e., pole resection) plus mesial resection (TPR+). The extent of lateral removal in the lobectomy types was applied to the same degree in each center for that resection type. The way to measure length of hippocampal resection was precisely defined and agreed upon between the three center surgeons. Therefore, the essential points of evaluation and surgical technique were agreed upon and outlined before the start of the trial. The histopathological diagnosis was done in the local depart-

ments of Neuropathology and the diagnosis and grade of hippocampal sclerosis were reviewed and confirmed by the reference center for histopathology of chronic epilepsy at the Department of Neuropathology in Erlangen.

Interventions The epileptologists in each center were free to decide whether they would propose a selective resection or one of the lobectomy types. The recommendation for type of resection (SAH or TLR) was based on presurgical evaluation and was followed strictly, independent of the study, and before patients were informed about the ongoing study, before the patients made their decision and, of course, also before randomization. The patients were then randomized into two groups concerning the intended length of resection of the hippocampus and parahippocampal gyrus. The only difference between the two groups was the extent of mesial resection regarding the hippocampal–parahippocampal bloc. The length of resection was to be determined intraoperatively after the opening of the temporal horn by using a narrow millimeter paper strip from the anterior tip of the temporal horn backwards placed on the hippocampal head along its length axis. Thus, mainly two large patient groups would be formed (SAH and TLR), each having been randomized into the 2.5- and 3.5-cm mesial resection subgroups. One group was planned to have a minimal resection of 25 mm determined in this fashion, the other group was to have a resection of a minimum of 35 mm. To simplify reading, the first group will henceforth be called the 2.5-cm group and the second the 3.5-cm group, with the implied understanding that this means the intended extent of resection. It was determined only at the beginning of surgery to which extent of resection a case should be allocated. The resection included major parts of the amygdaloid body in each case and the uncus. For the partial anterior one-third lobectomy, a polar resection of approximately 3 cm was done and for so-called anterior two-thirds temporal lobectomy, a resection of between 4.5 and 5 cm of the lateral neocortical temporal lobe was also included. The postoperative management was identical in both groups. Antiepileptic drugs were continued until the end of the observation period of 1 year after surgery, usually another year after that. The study design was reviewed and agreed on by the ethical boards in all three institutions.

Objectives, outcome measures, and sample size The objective of this study was to assess the significance of the extent of resection of mesial structures (hippocampus and parahippocampus) to achieve seizure freedom after surgery for temporal lobe epilepsy. Therefore, the following formal hypothesis was tested:

Postoperative seizure freedom is independent from the extent of resection.

Primary outcome measure was seizure freedom at 1 year and during the full first year expressed as class I in the Engel outcome scale. To increase the quality of follow-up, patients were seen postoperatively at 3 months, 6 months, and 1 year. We are well aware that later decline in seizure-free rate could theoretically affect the groups differently.

Sample size was determined to be able to detect the advantage of 80% seizure freedom for one resection type against a rate of seizure freedom of 60% for the other resection types with a power of 80% using a two-sided Fisher's exact test at a level of 5%. Thus, smaller effect sizes of 5% or 10% would not be detectable. A computer-generated blocked randomization list was prepared at the Institute of Medical Biometrics, Informatics and Epidemiology. Each center received the randomization codes in numbered and sealed envelopes, each envelope allocating the extent of resection for one particular patient in that center. The envelopes were to be used in the given sequence for the patients as they were included into the study. They had to be opened in the OR not before the morning of surgery. Thus, the type of surgery (selective amygdalohippocampectomy or lobectomy) was determined before the envelope was opened, but the content of the envelope determined the mesial extent of resection. Study participants were assigned to their resection extent group by that randomized sequence independent from the physicians involved with presurgical evaluation and recruitment of patients.

Neither the study participants nor the epileptologists assessing the outcomes were told of the result of the randomization, i.e., they were not aware whether a short or a long resection had been done; the only person knowing this was the surgeon. OR notes did not mention the planned resection extent. In particular, the persons performing the MRI volumetry were blinded to group assignment.

The primary goal was the comparison of rates of continuous seizure freedom between the two groups with intended 2.5-cm and 3.5-cm resection during 1 year after surgery. The outcome applies to the total population; all types of surgery were lumped. However, another study addressing just one operation type could come to different conclusions.

The rates were compared using the Fisher's exact test primarily for the population of patients who had followed the study protocol for at least 12 months and for whom the primary outcome could be measured. That is, follow-up was identical for all centers and all resection types and intended resection intent groups.

A secondary analysis was performed taking into account all randomized patients whereas drop-outs, severe violations of the protocol, and the impossibility to measure the outcome were counted as failure of therapy (intention-to-treat analysis). For both analyses, the patients were analyzed as part of the group to which they were

randomized. More analyses were performed based on the per-protocol population comparing groups of patients defined by the true extent of resection. This is a smaller group ($n=179$) defined as those with complete volumetric data. A separate publication is devoted to these secondary analyses based not on intended resection extent but on truly measured resection extent. Postoperative volumetric analyses were used in this study only to test whether the randomization into 2.5- and 3.5-cm groups did in fact lead to different resection volumes for the two groups.

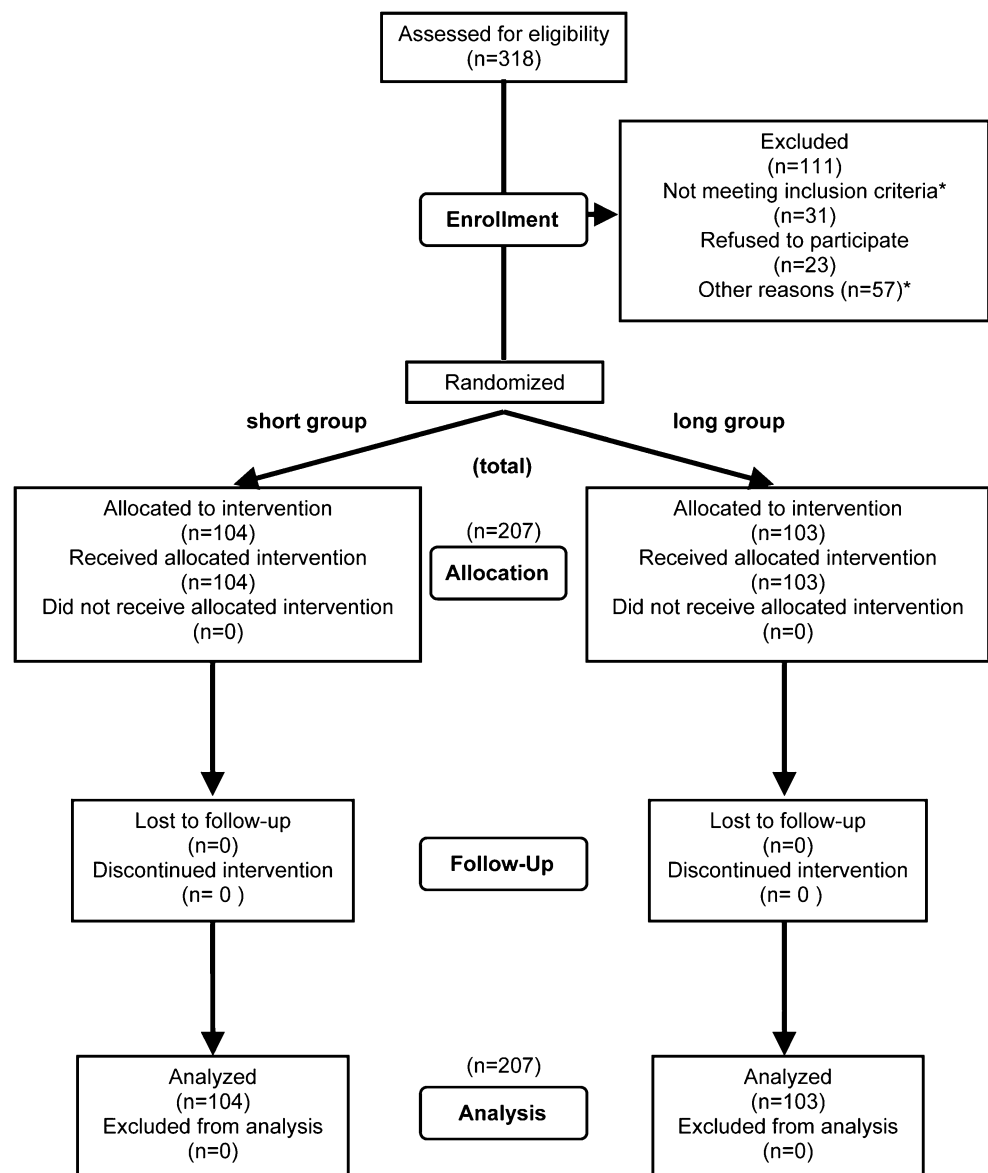
Results

A total of 318 cases were assessed for eligibility because of temporal lobe epilepsy surgery. Thirty-one of these did not meet inclusion criteria, 23 refused to participate, and 57 were excluded for other reasons (Fig. 1). Of the 31 not meeting inclusion criteria, the reasons were: inability to do proper neuropsychological testing because of retardation or foreign language ($n=16$), previous temporal lobe surgery ($n=4$), seizure duration less than 2 years ($n=1$), age below 18 years ($n=4$), and mesial resection restricted to uncus and amygdala ($n=6$). The 57 other exclusion reasons were: no immediate preoperative MRI because of adipose body size, MRI breakdown or missing MRI ($n=4$), forgotten randomization or envelope opened too late ($n=7$), and primarily no randomization because of pathology necessitating a dorsal reaching resection not allowing for randomization ($n=46$).

A total of 207 patients were allocated to the two intervention types, 103 to the 3.5-cm group and 104 to the 2.5-cm group. A total of 207 patients received the allocated intervention. Two patients died, one randomized to the short group by suicide 13 months after his operation with outcome Engel I. The second death (traffic accident) occurred 9 months after the operation, but we have a 7 months' outcome available for him and carried this forward for the primary analysis. The group available for the primary analysis consisted of these 207 cases. For the primary analysis, only the 1 year outcome and the result of the randomization procedure had to be available.

Baseline data Of 207 analyzed patients, 49 were recruited in center 1, 56 in center 2, and 102 in center 3. A total of 125 SAHs were performed, 53 anterior one-third lobectomies, and 15 standard two-third temporal lobectomies, four mesial lesionectomies plus hippocampo-parahippocampectomy, two tailored resections. Gender distribution, mean age at time of operation, duration of epilepsy, side of surgery, and age at onset of seizure disorder are shown for all analyzed patients, for the 2.5-cm and 3.5-cm resection group and the

Fig. 1 Patient flowchart detailing enrollment, allocation, and groups with follow-up and analyzed as proposed by the international randomized controlled trials registry



*Reasons explained in text

subgroup with mesial temporal (hippocampal) sclerosis. There were no statistically significant differences between the groups concerning all these variables. These baseline data for the three centers were tested for center effects, but no center effects could be detected (Tables 1 and 2).

The following risk factors were found in the past history: meningitis/encephalitis ($n=36$), trauma ($n=20$), febrile seizures ($n=37$), hypoxia ($n=3$), and prematurity ($n=3$). There were no differences in frequency of these risk factors between the 2.5-cm and 3.5-cm resection groups.

The total patient cohort ($n=207$) consisting of 104 with 2.5-cm and 103 patients with 3.5-cm resection was analyzed according to protocol. Histology in the 21 patients without MTS was: ganglioglioma WHO I ($n=5$), focal cortical dysplasia ($n=6$), gliosis ($n=4$), cavernoma ($n=2$), no

neuropathological alteration ($n=2$), microdysgenesis ($n=1$), and hamartomatous malformation ($n=1$).

Outcomes The intent to create two groups with a different extent of resection was successful as demonstrated in Fig. 2 where the box plot shows a significant difference ($p=0.001$) in mean true resected hippocampal volume between the group with intended 2.5-cm and intended 3.5-cm resection. If the resected volume of *combined* hippocampus and parahippocampal gyrus was considered, the difference between the groups was also statistically significant ($p=0.01$). The extent of resection was also different if not expressed as percent resected volume but as resected volume in cubic millimeters or if expressed as length of resection. It could therefore be demonstrated that the

Table 1 Baseline demographics

	HC resection length		Subgroups		Total analyzed patients (n=207)		
			True extent of HC resection volume (%)		OP type		
	2.5 cm (n=104)	3.5 cm (n=103)	MTS (n=186)	25% of cases with smallest resection	25% of cases with largest resection	Selective AHE (n=125)	Lobectomy [2/3 temp. resection and ant. lobectomy] (n=74)
Gender (female/male)	51/53	56/47	97/89	23/22	30/15	59/66	43/31
Mean age at timepoint of surgery	39.5±13.9	39.8±12.5	39.5±13.2	38.7±12.8	37.6±13.4	40.7±12.5	38.5±13.9
Side of pathology (left/right)	57/47	53/50	103/83	17/28	22/23	67/58	37/37
Duration of epilepsy (years, median)	22.5±14.0	21±14.3	21.5±13.8	24±12.5	18±14.0	24±13.8	22±14.9
Age of onset of seizure disorder (years, median)	11±10.8	14±11.4	12±11.5	10±10.4	13±12.2	13±11.4	10±10.90
No significant differences were found between the groups concerning the variables sex, age, side of pathology, and duration of epilepsy							

surgical technique and the randomization successfully achieved to create two subgroups with clearly different mean resection volumes. Randomization was also successful in that no differences were seen between the two resection groups and variables sex, age, age at onset of epilepsy, and duration of epilepsy. Figure 2 also demonstrates the range of results around the median value of truly resected volume.

Seizure freedom defined as Engel outcome class I was seen in 73.4% of the total patient group. Seizure outcome classes Engel I–IV were not different for the group with intended 2.5-cm versus intended 3.5-cm resection (Fig. 3). There was no significant difference for the frequency of class I seizure outcome comparing the 2.5-cm and 3.5-cm resection group with 74% and 72.8% Engel class I ($p=0.843$, Table 3), respectively. Similarly, there was no difference in outcome using the ILAE classification: 65.4% for 2.5-cm versus 69.9% for 3.5-cm group ($p=0.800$). This was also true whether looking at parahippocampus (PHG) resection volumes alone or at combined PHG plus hippocampus (HC) volumes. The significantly different true resection volume for the 2.5-cm and 3.5-cm group is consistently found when looking at the seizure-free and non-seizure-free subgroups (Fig. 4). Remarkably, the non-seizure-free patients did not show a smaller median resection extent for the group.

A similar analysis for the same groups mentioned in Table 3 based on resection *volume*, not on resection *length* as in Table 3 also failed to show significant differences in frequency of class I outcome. Comparable statistical analysis using extent of resection *length* in absolute numbers had shown similar results: no significant differences in resection length for seizure-free versus non-seizure-free patients and statistically significantly different resection lengths for the intended 3.5-cm resection groups versus the intended 2.5-cm resection groups.

There were no significant differences between the 2.5-cm and 3.5-cm resection group concerning neurological complications ($p<0.605$), visual field defects ($p<0.856$), or surgical complications ($p<0.875$). With just below 3% neurological complications and a permanent morbidity of 1.67%, these were a little lower than in the previous published range [4, 9]

Discussion

Overall evidence

The primary analysis of all randomized patients clearly shows that seizure freedom rate for the intended 2.5-cm resection is not statistically significantly different from the

Table 2 Patient characteristics in the three centers

Center	n (%)	Sex (female/male)	Mean age	Side of Pathology (left/right)	Duration of epilepsy (years, median)	Age at onset of epilepsy (years, median)
Center 1	49 (23.7)	29/20	39.2±13.2	27/22	21±15.2	12±11.4
Center 2	56 (27.1)	31/25	37.9±12.8	27/29	19±14.1	9±12.2
Center3	102 (49.3)	47/55	40.8±13.5	56/46	26.0±13.7	13.3±10.5
Σ	207	92/87	39.5±12.9	94/85	21.5±13.8	13±11.4

No significant differences were found between the centers concerning the variables sex, age, side of pathology, and duration of epilepsy

seizure freedom rate for the far posterior reaching larger 3.5-cm resection type.

The study at first sight is not in agreement with the only other study that prospectively randomized for extent of

mesial resection [42]. A total of 13 studies—nearly all non-randomized, several not volumetrically assessed—looked at extent of resection as a seizure outcome factor [33]. Only five of these [6, 7, 28, 35, 42] described that larger extents of resection led to better outcome, although Bonilha et al. (2007) [7] found this only for extent of hippocampus and entorhinal cortex resection, not for overall size of resection, and Siegel et al. (1990) [35] found this only for extent of parahippocampal gyrus resection. Eight studies could not confirm that extent of resection led to better outcome. Jack et al. (1988) [16] found that total amount of removal not

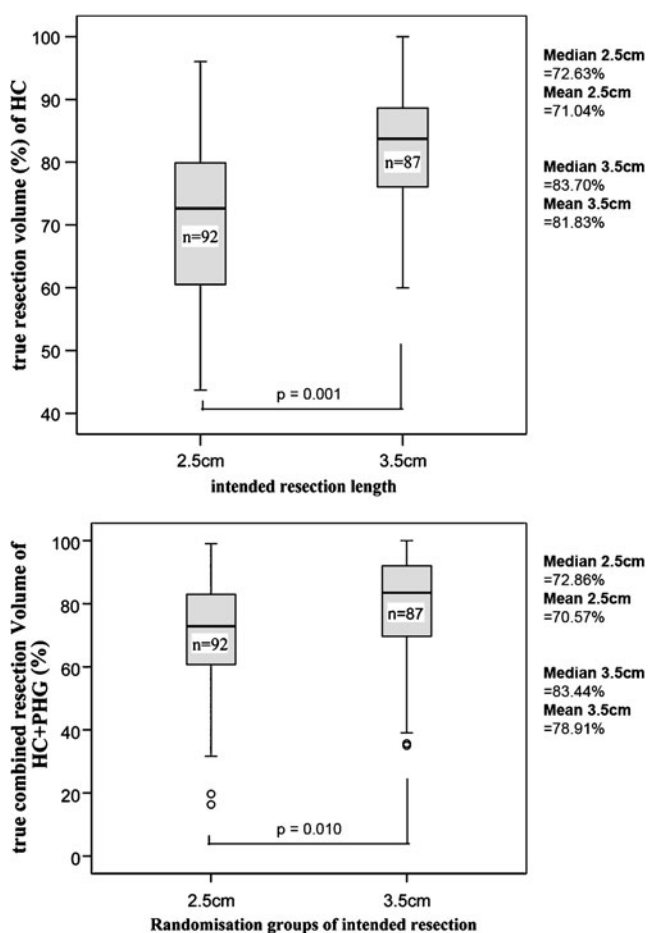


Fig. 2 True HC and combined HC–PHG resection volume (%) versus intended resection length. The data show secondary analysis for the group with available volumetric data ($n=179$). These box plots compare the true resection extent of hippocampus (HC) in the upper part and of combined hippocampus and parahippocampus (PHG) in the lower part. The truly achieved resection volumes are shown for the two groups with intended resection length of 2.5 versus 3.5 cm. For both comparisons, considering HC alone or combined HC and PHG, the median true resected volume for the two groups with the 2.5-cm and 3.5-cm intended resection extents is significantly different

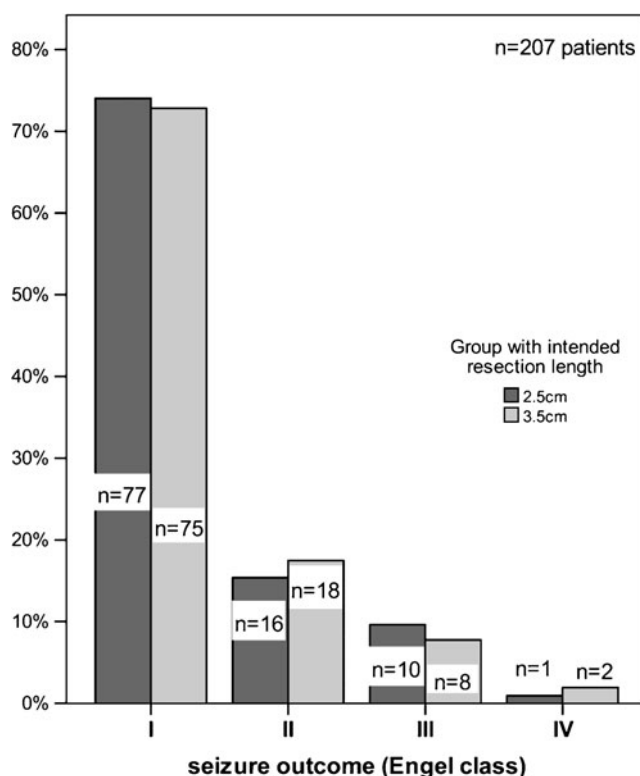


Fig. 3 Seizure outcome classes for 2.5-cm versus 3.5-cm intended resection group. The data show primary analysis of all randomized patients ($n=207$). The bar graph shows the comparison of the outcome in the two groups with an intended resection length of 2.5 and 3.5 cm subdivided into Engel outcome class I–IV. The length of the bar describes the percentage in that seizure outcome class and the numbers at the bars are the absolute patient numbers

Table 3 Intent-to-treat analysis of seizure freedom rates (Engel outcome class I)

Intent-to-treat		Total <i>n</i>	Engel class I outcome		<i>p</i>
			<i>n</i>	% of total <i>n</i>	
All		207	152	73.4	–
HC resection length, long vs. short group	2.5 cm	104	77	74	0.843
	3.5 cm	103	75	72.8	

necessarily correlated with outcome. Jones-Gotman et al. (1997) [17] found that significantly different extents of removal of hippocampus and amygdala were compatible with seizure freedom. Renowden et al. (1995) [32] found a similar seizure outcome for hippocampal resections of less than 2 cm or more than 2 cm and generally that extent of removal of amygdalum and hippocampus was not directly related to seizure outcome. Van Rijckevorsel et al. (2005) [38] found that extent of hippocampal removal in SAH did not correlate with seizure outcome. Similar findings described that widely variable extents of hippocampal removal obtained by tailoring with intraoperative recordings were compatible with seizure freedom [2, 16, 18, 24, 36].

A detailed analysis based on really achieved extent of resection for $n=179$ and other analyses performed *outside the randomization frame* for the whole group ($n=207$) will look at variability of resection extent, subgroups, possible confounding factors, and possible differences between the lobectomy and SAH groups, but will be published in a separate paper (Schramm 2010 et al.) [34].

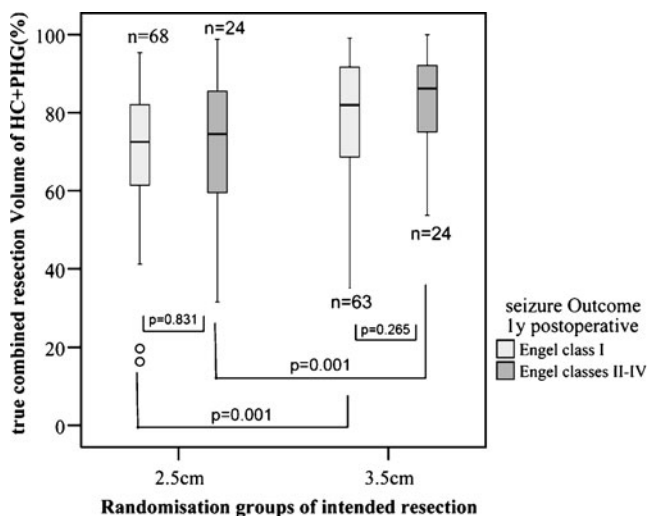


Fig. 4 True combined HC–PHG resection volumes in the 2.5-cm and 3.5-cm groups shown separately for patients with Engel class I outcome and combined Engel class II–IV outcomes. The box plots confirm that true median resection extents for the 2.5-cm and 3.5-cm groups are different, now demonstrating this for the seizure-free and non-seizure-free subgroups. At the same time, true median resection extents do not differ for seizure-free and non-seizure-free subgroups

Methodology

The choice of 2.5 cm and 3.5 cm for the two treatment groups was a pragmatic one: it was close to values around 2 cm from the literature and it was necessary to have an easy intraoperative measure available. It had to be a measure practical in the intraoperative set up and easy to do, which would not necessarily have been the case for resection extents based on anatomical landmarks such as “to the level of posterior border of brain stem” or in relation to the choroidal point such as “10 mm behind the choroidal point” versus “20 mm behind the choroidal point”. We were well aware of the fact that anatomic variation, the degree of hippocampal atrophy, the degree of temporal lobe atrophy, and variability in enlargement of the temporal horn would influence this measure.

It should be noted that the study design did not aim to compare a *minimal* with a *large* hippocampal resection. The intention was to compare an easy to perform well-sized resection with a large hippocampal resection. The similarity in seizure outcome for the intended smaller resection compared to the intended larger resection should not lead to the wrong conclusion that the amount of hippocampal removal is completely irrelevant.

Due to the prerequisite of creating two groups with a shorter and longer resection, cases had to be excluded where only a lesionectomy in the uncus or amygdalum was to be performed, but more frequently cases could not be included where due to the results of presurgical evaluation, a far posteriorly reaching resection was advised.

We do not consider the fact that lobectomies and SAHs, i.e., different resection types, were included in the study to be a disadvantage or to be able to devalue the results. It is meaningful to only vary the extent of mesial resection as has been done before by Wyler et al. (1995) [42] or to only vary the extent of lateral resection, which has also been done in a prospective randomized study [15]. The fact that TLR and SAH were both acceptable resection types in the study design reflects the reality of TLE surgery worldwide. Some epileptologists continue to prefer a variant of the TLR and others feel that a more mesial restricted resection type like SAH is possible in many TLE cases. Two of the three centers employed both TLR and SAH resection types

and a third center always preferred to do an anterior lobectomy. So, whereas in one center anterior lobectomy constituted 100% of all resections, in the two other centers, SAH constituted 73% and 82% of resections. More restricted resection types such as mesial lesionectomy or tailored resection were done in one center in 5.4% and in the other center in 4.9% of cases. Thus, the choice of resection types leads to two large patient groups with SAH and lobectomies. We feel this reflects well a certain trend mainly present in Europe to go away from standardized lobectomy. Whether the preference for one resection type in a specific center should be called a bias or whether this justifiably reflects the individual conviction of the decision making of the epileptologist based on the results of presurgical evaluation and his interpretation of the world literature is left for the reader to decide.

Center effects from patient characteristics appear not very likely since the demographic parameters of the patients were not significantly different between the centers. In particular, this appears to be true for several factors known to be outcome variables for seizure freedom after temporal lobe surgery, such as duration of epilepsy or presence of hippocampal sclerosis. Other center-dependent effects, however, are possible.

Limitations

The results of this study cannot be generalized for each and every type or of temporal lobe resection; in particular, they can only be applied for cases where a mesial resection was recommended which was not a priori decided to reach to the very posterior aspect of the temporal lobe mesially. Other limitations are that the results are not applicable for patients under 18 years of age. Another possible limitation is that the development after the 1-year period of follow-up available for this study may differently affect the resection groups with 2.5 cm and 3.5 cm. Since the study was powered to detect a difference of 20%, smaller effect sizes of 5% or 10% may not be detectable. However, these results can be used for the vast majority of temporal lobe resections which include an extent of mesial resection of the hippocampus–parahippocampus area classically used in a typical SAH procedure or partial lobectomy procedure supplemented by mesial resection.

The study hypothesis was that postoperative seizure freedom is independent from the extent of resection. In this study, we found no significant differences in seizure freedom between the two resection extent groups using the intent-to-treat analysis.

This is in contrast to the study of Wyler et al. (1995) [42]. Wyler's cases were all temporal lobectomies and only the mesial extent of resection was randomly assigned to his

patients creating one group with a *shorter* resection (i.e., ...“resected only to the level of the anterior margin of the cerebral peduncle”) and one with a *maximal* resection (i.e.,...“more posteriorly to the level of the superior colliculus”). That study design constitutes a difference to the one from this study, where a “well-sized” resection (i.e., intended 2.5 cm) is compared with a large (i.e., intended 3.5 cm) resection. Maybe, the short length of Wyler's smaller resection is an explanation for the large difference in percent seizure freedom between his two groups, 38% seizure-free for the small and 69% for the large resection group. In contrast, the smaller resection in this study removed 72.6% of the hippocampus volume as a median value for the whole 2.5-cm group. It appears that the study designs were rather different with Wyler's small resection group aiming for a rather small resection volume and the larger group aiming for a maximal volume. In the present study, the difference between the two resection groups being unremarkable is possibly because the smaller resection group already removed over 72% of hippocampus volume as a median for the group. One might say that the Wyler study proved that a small resection leads to much poorer results than a truly maximal resection (at least by design) whereas this study demonstrated that maximizing the resection extent did not further improve seizure outcome. Wyler's study design also differed in so far as the extent of lateral lobe removal was planned to be identical whereas in this study, it varied in the TLR group. An important difference between Wyler's randomized study and this study is the assessment of true resection extent by MRI volumetry in the latter one. An amazing amount of variability in extent of resection (as demonstrated in Figs. 2 and 4) was found in this study and in all others that ever published volumetric results. It remains unknown for Wyler's randomized study, and one could argue that it is not so important to know since being an intent-to-treat study, results should only be evaluated per intent-to-treat. Acknowledging the request of several authors [15, 23, 25] for postoperative volumetric assessment of resection extent, we feel that the results of the previous study [42] with randomized mesial resection extent are less precisely supported than results from this study.

If discussing the necessary amount of hippocampal–parahippocampal removal, various schools of thought might be followed: (1) a small mesial resection is sufficient, (2) a maximal resection is best, and (3) the amount of HC–PHG removal needs to be sufficiently large enough to disturb the build-up and maintenance of a seizure event, but does not have to be maximal to achieve this. The results of this study, where maximizing mesial resection extent did not lead to an increase in seizure freedom, lead to the concept that it is not *maximal* volume resection that may

help to achieve optimal seizure freedom but the resection of an *adequate* volume of resection in the fitting area. Similar ideas have been phrased by Abosch et al. [1] who suggested that “...there is a critical volume of amygdala, hippocampus, and/or entorhinal cortex that must be resected to obtain optimal seizure control...”.

Another argument that not maximal resection is the most important factor for seizure outcome can be derived from the fact that the many different centers worldwide doing temporal lobe resections achieve similar seizure freedom results despite various resection techniques and including centers who do not aim to achieve maximal mesial resection. How could it be possible that a sub-maximal extent of resection can achieve the same percentage of seizure freedom as a bigger one. Maybe, the reason is that enough fiber tracts are interrupted by a well-sized or normal-sized resection to inhibit the build-up and propagation of epileptic discharges. Abosch et al. (2002) [1] have proposed that “...it is possible.....that it is the percentage of afferent and efferent connections.....and not the volumes of the structures themselves, that is the critical factor in determining seizure outcome...”. The latter argument has been used in the past in the discussion of cases with temporal resection after initial failure, where a considerable proportion of re-operated patients did not become seizure-free despite maximal resection, for example by Germano et al. [11], and discussed in some detail in a recent review [33].

Acknowledgements This study was funded as part of the transregional collaborative research consortium SFB-TR3 “Mesial temporal lobe epilepsies” of the Deutsche Forschungsgemeinschaft (DFG). The authors thank H. Urbach M.D. and R. Koenig M.D. (Department of Neuroradiology, University Clinic, Bonn) as well as I. Bluemcke M.D. (Department of Neuropathology, University Clinic, Erlangen) for help with MRI volumetry and reference pathology. Additional thanks go to other members of the study group: S. Roeske, S. Kaaden, C. Scheiwe, M.D., B. Harzheim, P. Süßmann, S. Ferl, and D. Meinken-Jäggi for organizational and volumetric support. K. Wagner, C. Helmstaedter PhD, S. Kaaden, S. Roeske, and F. Oltmanns helped with neuropsychological assessment, B.J. Steinhoff M.D., M. Merschhemke M.D., C. Dehnicke M.D., and C. Bien, M.D. helped with presurgical evaluation. D. Haun PhD helped with statistics and graphs.

Conflicts of interest None.

References

- Abosch A, Bernasconi N, Boling W, Jones-Gotman M, Poulin N, Dubeau F, Andermann F, Olivier A (2002) Factors predictive of suboptimal seizure control following selective amygdalohippocampectomy. *J Neurosurg* 97:1142–1151
- Arruda F, Cendes F, Andermann F, Dubeau F, Villemure JG, Jones-Gotman M, Poulin N, Arnold DL, Olivier A (1996) Mesial atrophy and outcome after amygdalohippocampectomy or temporal lobe removal. *Ann Neurol* 40:446–450. doi:10.1002/ana.410400314
- Awad IA, Katz A, Hahn JF, Kong AK, Ahl J, Luders H (1989) Extent of resection in temporal lobectomy for epilepsy. I. Interobserver analysis and correlation with seizure outcome. *Epilepsia* 30:756–762
- Behrens E, Schramm J, Zentner J, König R (1997) Surgical and neurological complications in a series of 708 epilepsy surgery procedures. *Neurosurgery* 41:1–9, discussion 9–10
- Blumcke I, Pauli E, Clusmann H, Schramm J, Becker A, Elger C, Merschhemke M, Meencke HJ, Lehmann T, von Deimling A, Scheiwe C, Zentner J, Volk B, Romstock J, Stefan H, Hildebrandt M (2007) A new clinico-pathological classification system for mesial temporal sclerosis. *Acta Neuropathol* 113:235–244. doi:10.1007/s00401-006-0187-0
- Bonilha L, Kobayashi E, Mattos JP, Honorato DC, Li LM, Cendes F (2004) Value of extent of hippocampal resection in the surgical treatment of temporal lobe epilepsy. *Arq Neuropsiquiatr* 62:15–20.
- Bonilha L, Yasuda CL, Rorden C, Li LM, Tedeschi H, de Oliveira E, Cendes F (2007) Does resection of the medial temporal lobe improve the outcome of temporal lobe epilepsy surgery? *Epilepsia* 48:571–578
- Clusmann H, Kral T, Schramm J (2006) Present practice and perspective of evaluation and surgery for temporal lobe epilepsy. *Zentralbl Neurochir* 67:165–182. doi:10.1055/s-2006-942144
- Clusmann H, Schramm J, Kral T, Helmstaedter C, Ostertun B, Fimmers R, Haun D, Elger CE (2002) Prognostic factors and outcome after different types of resection for temporal lobe epilepsy. *J Neurosurg* 97:1131–1141. doi:10.3171/jns.2002.97.5.1131
- Feindel W, Rasmussen T (1991) Temporal lobectomy with amygdaloectomy and minimal hippocampal resection: review of 100 cases. *Can J Neurol Sci* 18:603–605
- Germano IM, Poulin N, Olivier A (1994) Reoperation for recurrent temporal lobe epilepsy. *J Neurosurg* 81:31–36
- Goldring S, Edwards I, Harding GW, Bernardo KL (1992) Results of anterior temporal lobectomy that spares the amygdala in patients with complex partial seizures. *J Neurosurg* 77:185–193
- Hadar E, Bingaman W, Foldvary M (2001) Prospective analysis of outcome after selective amygdalohippocampectomy and anterior temporal lobectomy for refractory epilepsy. Congress of Neurological Surgeons, San Diego
- Helmstaedter C, Richter S, Roske S, Oltmanns F, Schramm J, Lehmann TN (2008) Differential effects of temporal pole resection with amygdalohippocampectomy versus selective amygdalohippocampectomy on material-specific memory in patients with mesial temporal lobe epilepsy. *Epilepsia* 49:88–97. doi:10.1111/j.1528-1167.2007.01386.x
- Hermann B, Davies K, Foley K, Bell B (1999) Visual confrontation naming outcome after standard left anterior temporal lobectomy with sparing versus resection of the superior temporal gyrus: a randomized prospective clinical trial. *Epilepsia* 40:1070–1076
- Jack CR Jr, Sharbrough FW, Marsh WR (1988) Use of MR imaging for quantitative evaluation of resection for temporal lobe epilepsy. *Radiology* 169:463–468
- Jones-Gotman M, Zatorre RJ, Olivier A, Andermann F, Cendes F, Staunton H, McMackin D, Siegel AM, Wieser HG (1997) Learning and retention of words and designs following excision from medial or lateral temporal-lobe structures. *Neuropsychologia* 35:963–973
- Kanner AM, Kaydanova Y, deToledo-Morrell L, Morrell F, Smith MC, Bergen D, Pierre-Louis SJ, Ristanovic R (1995) Tailored anterior temporal lobectomy. Relation between extent of resection of mesial structures and postsurgical seizure outcome. *Arch Neurol* 52:173–178
- Keogan M, McMackin D, Peng S, Phillips J, Burke T, Murphy S, Farrell M, Staunton H (1992) Temporal neocorticectomy in

- management of intractable epilepsy: long-term outcome and predictive factors. *Epilepsia* 33:852–861
20. Leonard G (1991) Temporal lobe surgery for epilepsy: neuropsychological variables related to surgical outcome. *Can J Neurol Sci* 18:593–597
 21. Loring DW, Lee GP, Meador KJ, Smith JR, Martin RC, Ackell AB, Flanigin HF (1991) Hippocampal contribution to verbal recent memory following dominant-hemisphere temporal lobectomy. *J Clin Exp Neuropsychol* 13:575–586
 22. Lutz MT, Clusmann H, Elger CE, Schramm J, Helmstaedter C (2004) Neuropsychological outcome after selective amygdalohippocampectomy with transsylvian versus transcortical approach: a randomized prospective clinical trial of surgery for temporal lobe epilepsy. *Epilepsia* 45:809–816
 23. McIntosh AM, Wilson SJ, Berkovic SF (2001) Seizure outcome after temporal lobectomy: current research practice and findings. *Epilepsia* 42:1288–1307
 24. McKhann GM 2nd, Schoenfeld-McNeill J, Born DE, Haglund MM, Ojemann GA (2000) Intraoperative hippocampal electrocorticography to predict the extent of hippocampal resection in temporal lobe epilepsy surgery. *J Neurosurg* 93:44–52
 25. Meencke HJ, Meinken-Jaggi D, Lehmann TN, Schmitt FC (2008) Outcome changes during 10 year follow-up after epilepsy surgery. *Epilepsia* 49:304
 26. Merschhemke MB, Lehmann TN, Meencke HJ (2004) Outcome results after surgical treatment of temporal lobe epilepsies with hippocampal pathology of non-HS type. *Epilepsia* 45:339
 27. Mueller CA, Scorzin J, Koenig R, Urbach H, Fimmers R, Zentner J, Lehmann TN, Schramm J (2007) Comparison of manual tracing versus a semiautomatic radial measurement method in temporal lobe MRI volumetry for pharmacoresistant epilepsy. *Neuroradiology* 49:189–201. doi:10.1007/s00234-006-0171-3
 28. Nayel MH, Awad IA, Luders H (1991) Extent of mesiobasal resection determines outcome after temporal lobectomy for intractable complex partial seizures. *Neurosurgery* 29:55–60, discussion 60–51
 29. Niemeyer P (1958) The transventricular amygdalohippocampectomy in temporal lobe epilepsy. In: Baldwin M, Bailey P (eds) *Temporal lobe epilepsy*, 3rd edn. Charles C. Thomas, Springfield, pp 461–482
 30. Novak K, Czech T, Prayer D, Dietrich W, Serles W, Lehr S, Baumgartner C (2002) Individual variations in the sulcal anatomy of the basal temporal lobe and its relevance for epilepsy surgery: an anatomical study performed using magnetic resonance imaging. *J Neurosurg* 96:464–473
 31. Olivier A (1991) Relevance of removal of limbic structures in surgery for temporal lobe epilepsy. *Can J Neurol Sci* 18:628–635
 32. Renowden SA, Matkovic Z, Adams CB, Carpenter K, Oxbury S, Molyneux AJ, Anslow P, Oxbury J (1995) Selective amygdalohippocampectomy for hippocampal sclerosis: postoperative MR appearance. *AJNR Am J Neuroradiol* 16:1855–1861
 33. Schramm J (2008) Temporal lobe epilepsy surgery and the quest for optimal extent of resection: a review. *Epilepsia* 49:1296–1307
 34. Schramm J, Lehmann T, Zentner J, Mueller CA, Scorzin J, Fimmers R, Meencke HJ, Schulze-Bonhage A, Elger CE (2010) Randomized controlled trial of 2.5 cm versus 3.5 cm mesial temporal resection. Part 2: volumetric resection extent and subgroup analyses. *Acta Neurochirurgica*. doi:10.1007/s00701-010-0901-5
 35. Siegel AM, Wieser HG, Wichmann W, Yasargil GM (1990) Relationships between MR-imaged total amount of tissue removed, resection scores of specific mediobasal limbic subcompartments and clinical outcome following selective amygdalohippocampectomy. *Epilepsy Res* 6:56–65
 36. Son EI, Howard MA, Ojemann GA, Lettich E (1994) Comparing the extent of hippocampal removal to the outcome in terms of seizure control. *Stereotact Funct Neurosurg* 62:232–237
 37. Urbach H, Hattingen J, von Oertzen J, Luyken C, Clusmann H, Kral T, Kurthen M, Schramm J, Blumcke I, Schild HH (2004) MR imaging in the presurgical workup of patients with drug-resistant epilepsy. *AJNR Am J Neuroradiol* 25:919–926.
 38. Van Rijkceversel K, CMd G (2005) Selective amygdalohippocampectomy: seizure outcome in 26 consecutive cases compared to the amount of resection. *Epilepsia* 46:253–260
 39. Wiebe S (2003) Randomized controlled trials of epilepsy surgery. *Epilepsia* 44(Suppl 7):38–43
 40. Wiebe S, Blume WT, Girvin JP, Eliasziw M (2001) A randomized, controlled trial of surgery for temporal-lobe epilepsy. *N Engl J Med* 345:311–318
 41. Wiebe S, Jette N (2006) Randomized trials and collaborative research in epilepsy surgery: future directions. *Can J Neurol Sci* 33:365–371
 42. Wyler AR, Hermann BP, Somes G (1995) Extent of medial temporal resection on outcome from anterior temporal lobectomy: a randomized prospective study. *Neurosurgery* 37:982–990, discussion 990–981